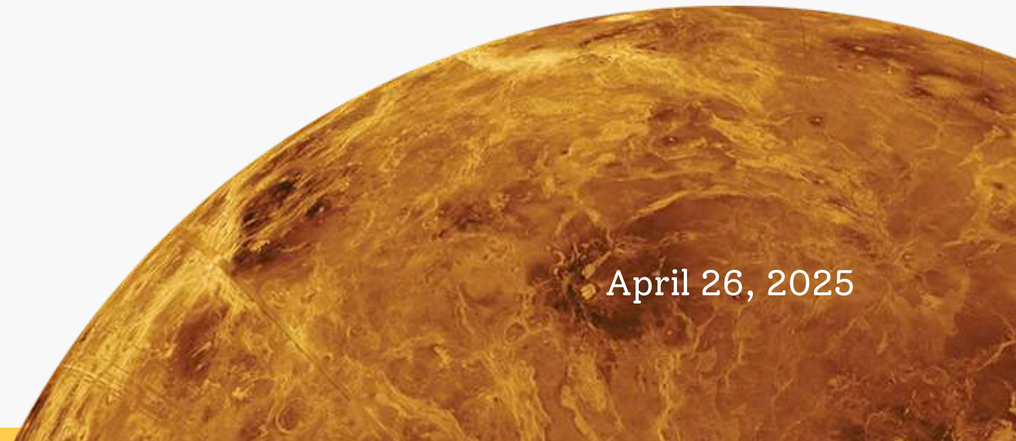
NASA L'SPACE Mission
Concept Academy

Team 27

THE VENERA NOVA MISSION

Isabela Benavidez, Rachel Brown, Victor Chen, Alaina Claudio, Andrew Hoang, Gregory Jabido, Elias Pedroza, Valerie Miranda, Isaac Munoz, Patrick Nguyen, Tyree Nguyen, Manahil Syeda



April 26, 2025

AGENDA OVERVIEW

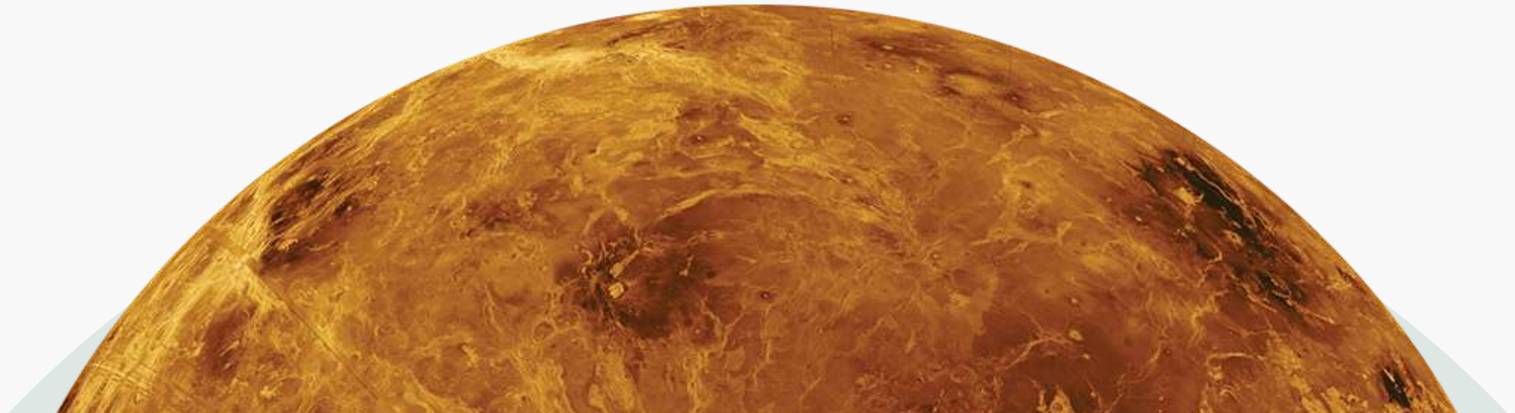
01 Introduction

02 Science

03 Engineering

04 Programmatics

05 Conclusion



TEAM INTRODUCTION

Tyree Nguyen



Deputy Project Manager of Resources

Victor Chen



Program Analyst / CDH Engineer

Isaac Munoz



Mission Assurance Specialist / Electrical Engineer

Rachel Brown



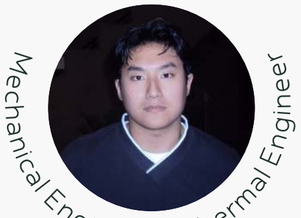
Project Manager

Manahil Syeda



Lead Systems Engineer

Patrick Nguyen



Mechanical Engineer / Thermal Engineer

Andrew Hoang



Electrical Engineer / Thermal Engineer

Gregory Jabido



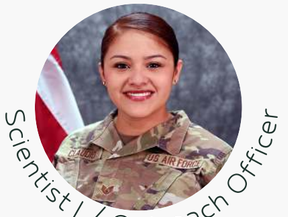
CDH Engineer / Electrical Engineer

Valerie Miranda



Chief Scientist

Alaina Claudio



Scientist I / Outreach Officer

Isabela Benavidez



Mechanical Engineer / Scientist III

Elias Pedroza



Thermal Engineer / Scientist II

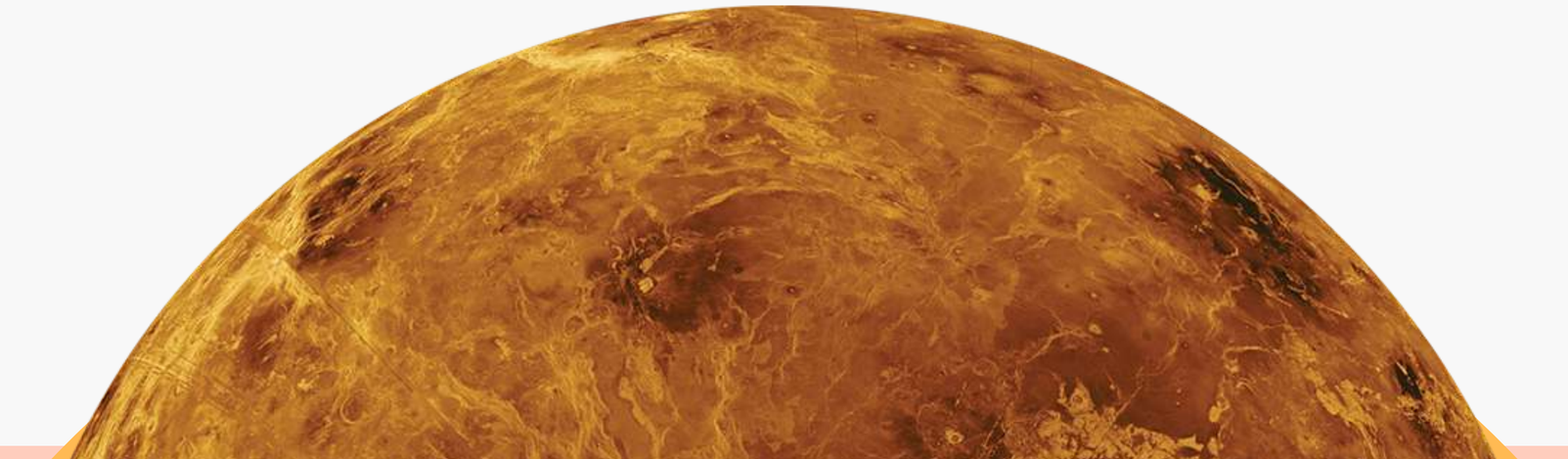
WHAT IS THE VENERA NOVA MISSION?

- Small, low-cost Discovery Mission to Venus
- Aerobot concept developed by team



Figure 1: Aerobot at NASA (JPL Robotics)

SCIENCE



MISSION STATEMENT

“The Venera Nova mission aims to investigate how Venus’s surface and interior influence and interact with its atmosphere by studying the planet’s superrotation, lack of magnetic field, and potential for life in the planet’s upper atmosphere.”

KEY CHARACTERISTICS:

- Superrotation
- Induced magnetic field
- Phosphine gas

SCIENCE TRACEABILITY MATRIX

Science Goals	Science Objectives	Science Measurement Requirements		Instrument Performance Requirements		Predicted Instrument Performance	Instrument	Mission Requirements
		Physical Parameters	Observables					
Determine how Venus' planetary surface and interior influence and interact with its atmosphere. “Decadal Question 6.4: How do planetary surfaces and interiors influence and interact with their host atmospheres?”	Determine the influence of Venus' superrotation on its surface and atmosphere.	Track cloud motion, brightness variations, and Kelvin waves every 10 seconds.	Capture images of Venus's clouds every 10 seconds in UV, visible, and near-infrared wavelengths. Collect wind speed data by tracking cloud motion every 10 seconds. Record periodic fluctuations in clouds.	Wavelength Range	365 - 1000 nm	365 - 1000 nm	Venus Monitoring Camera (VMC)	Mission shall measure cloud dynamics including wind speed and brightness variations through high-quality atmospheric data which will be obtained through the Aerobot's descent.
				Integration Time	0.5 ms - 30 s	10 s		
				Sensitivity	0.3 to 1.0 µm	0.3 to 1.0 µm		
				Spatial Resolution	0.2 - 45 km	~ 1.5 km		Mission shall measure atmospheric spectra, including ion composition and absorption features, as the Aerobot descends.
	Determine how a lack of magnetic field affects atmospheric composition.	Measure the composition, ionization, and the escape rates of particles and solar photons in the upper atmosphere.	Collect ionized species such as O+, H+, CO2+ in the UV range (118-320 nm) over a 1 km range every hour.	Wavelength Range	118 - 320 nm	118 - 320 nm	SPICAV (Ultraviolet Spectrometer)	
				Integration Time	80 - 450 ms at 1 km	80 - 450 ms at 1 km		
				Sensitivity	200 - 300 nm	200 - 300 nm		All science operations shall be completed within 48 hours after deployment of the EDI system.
				Spatial Resolution	100 km	100 km		
	Determine the probability of life existing within Venus' upper atmosphere.	Confirm presence of phosphine gas in the upper atmosphere.	Detect phosphine peaks between 190 and 220 nm.	Wavelength Range	2.2 to 4.3 µm	2.2 to 4.3 µm	SPICAV (Solar Occultation InfraRed Spectrometer - SOIR)	
				Integration Time	5 s	5 s		
				Sensitivity	2257–4430 cm−1	2257–4430 cm−1		Aerobot will descend from 70 - 50 km above the Ovda Regio region.
				Spatial Resolution	0.025-cm−1	0.025-cm−1		

MISSION LOCATION

- Ovda Regio, a plateau within a highland region named Aphrodite Terra

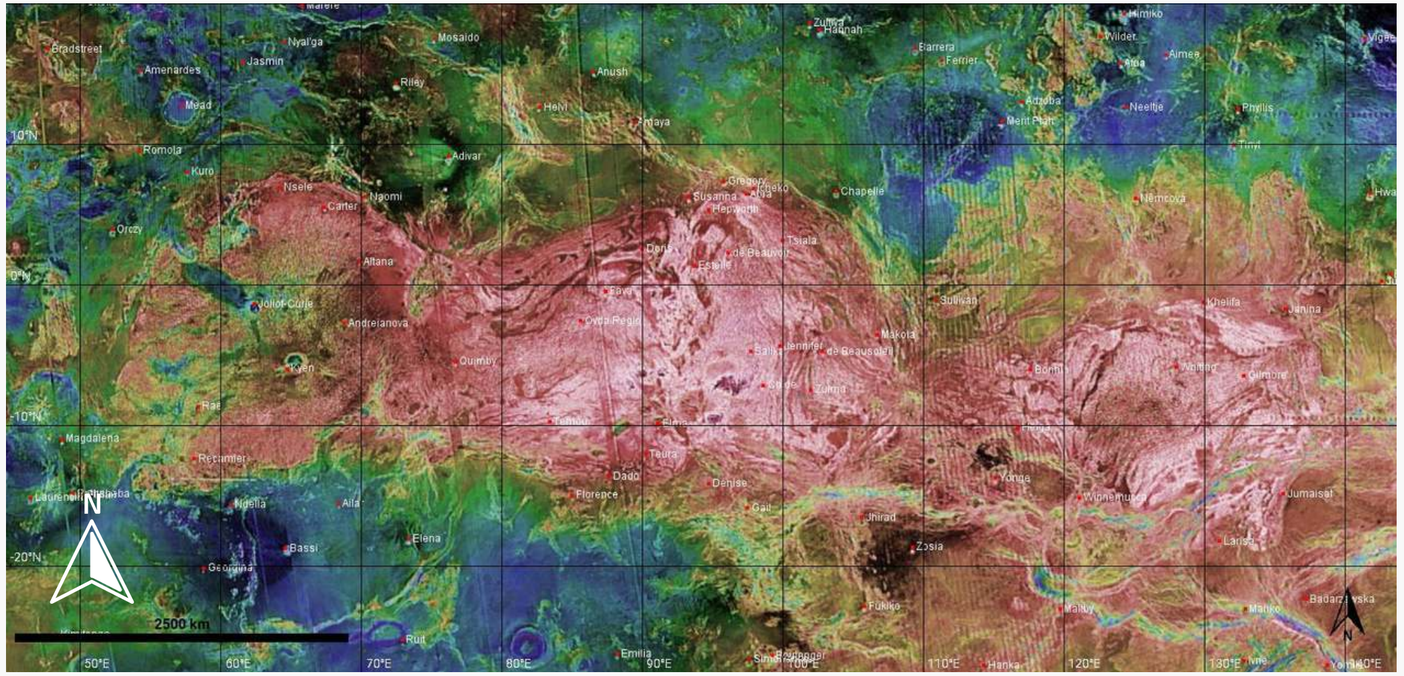


Figure 2: Image of Ovda Regio (JMARS)

MISSION LOCATION

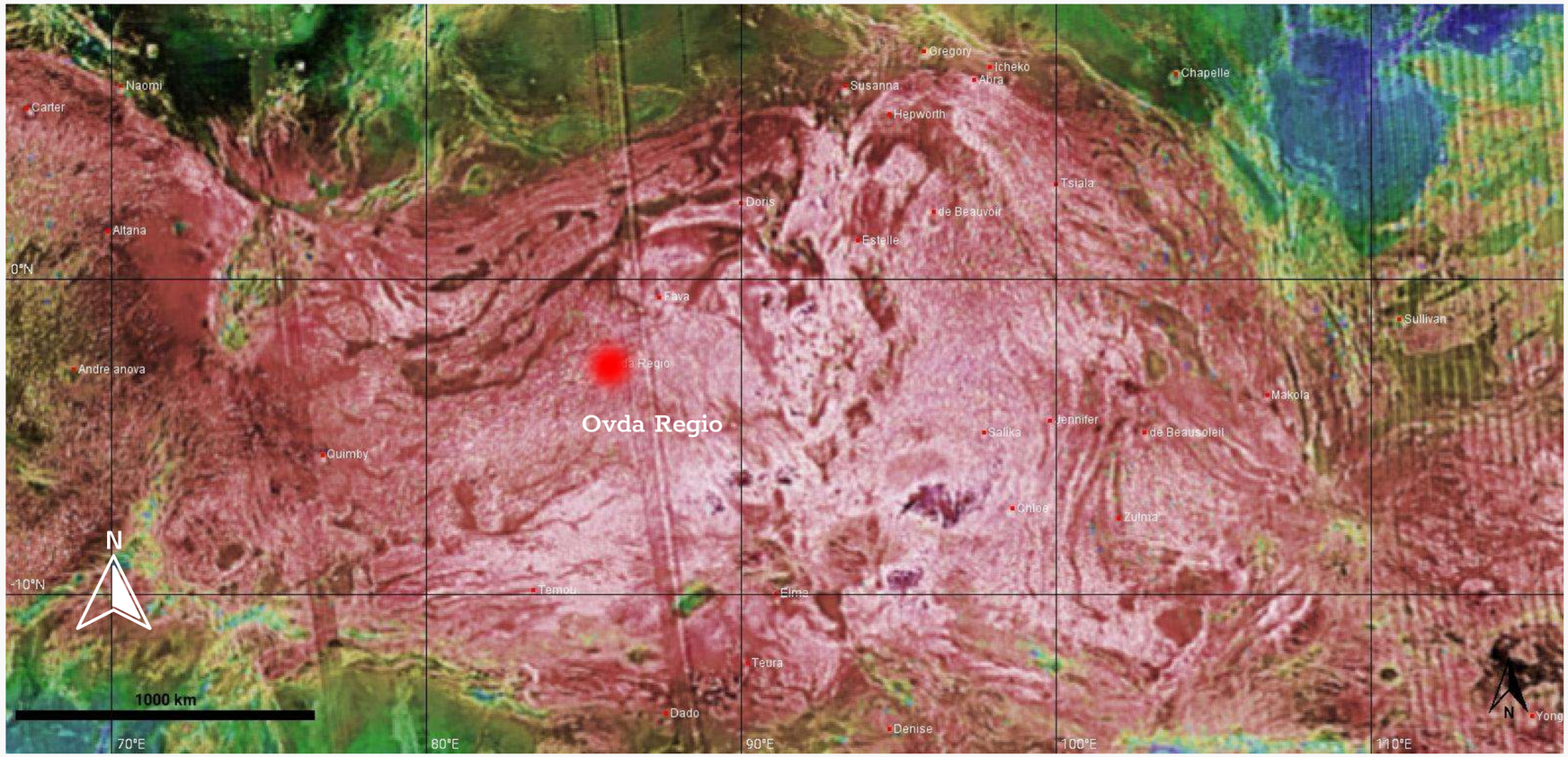


Figure 3: Zoomed in of Ovda Regio (JMARS)

CONOPS

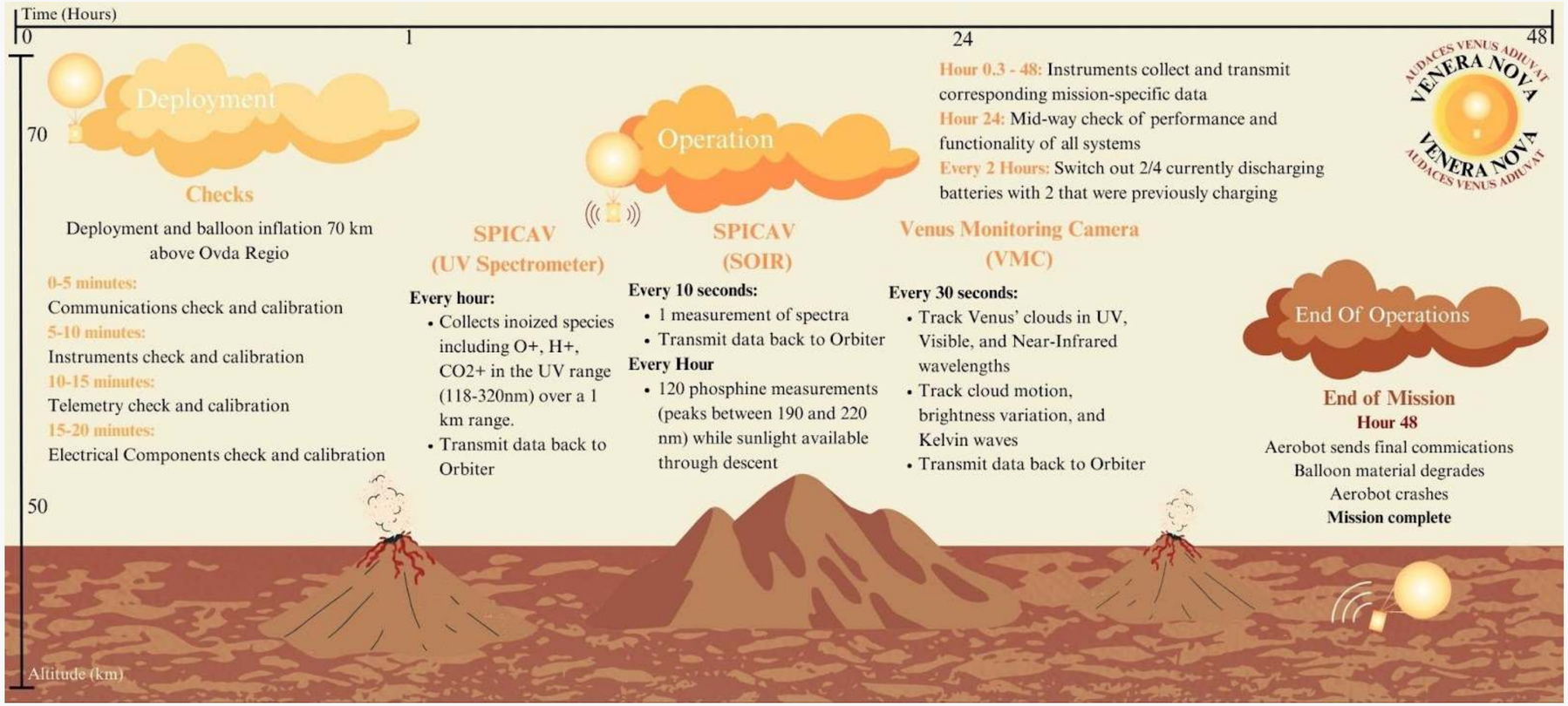
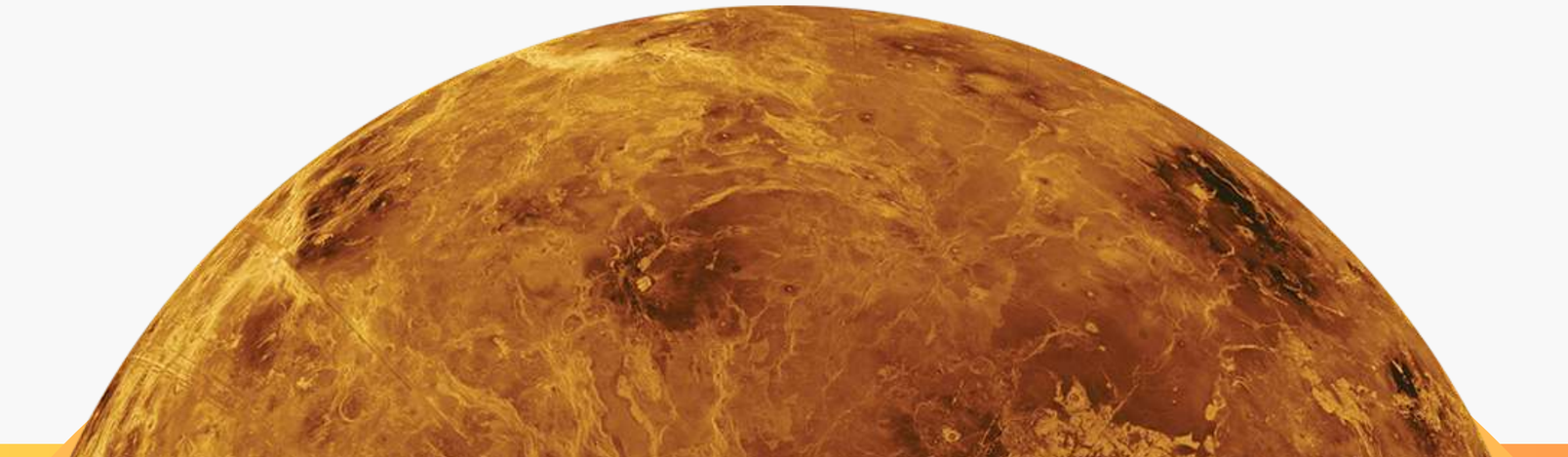


Figure 4: Concept of Operations for the Venera Nova Mission

ENGINEERING



ENGINEERING

MECHANICAL

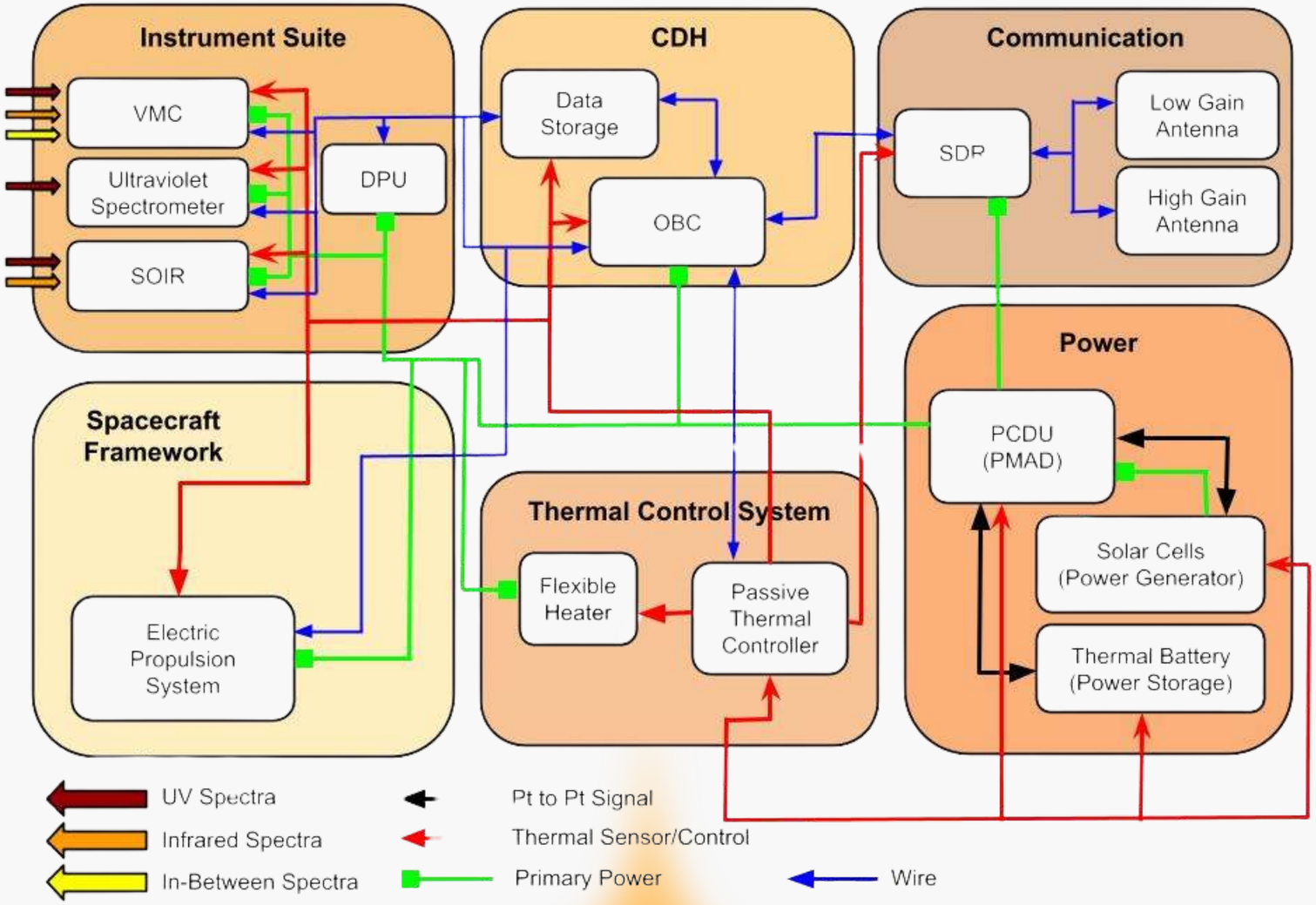
THERMAL

PAYLOAD

POWER

CDH

SUBSYSTEM OVERVIEW



PAYLOAD SUBSYSTEM

UV SPECTROMETER

SOLAR OCCULTATION INFRARED SPECTROMETER (SOIR)

VENUS MONITORING CAMERA (VMC)

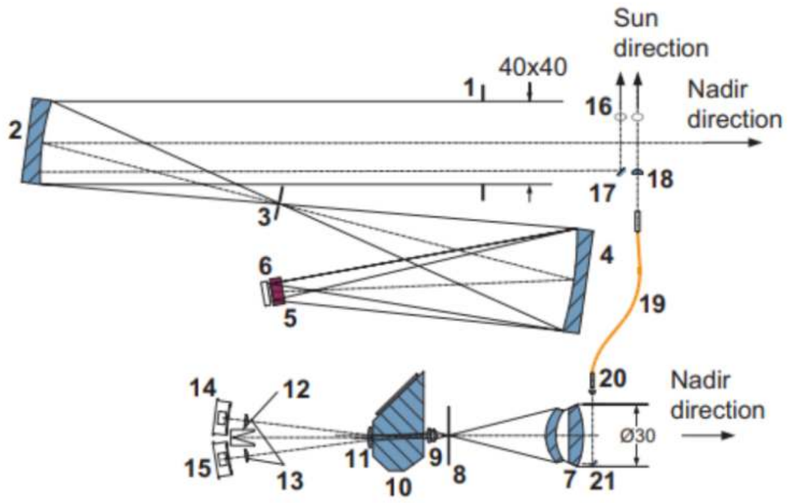


Figure 5: Optical scheme of the UV and IR channels of SPICAV (Bertaux et al., 2007)

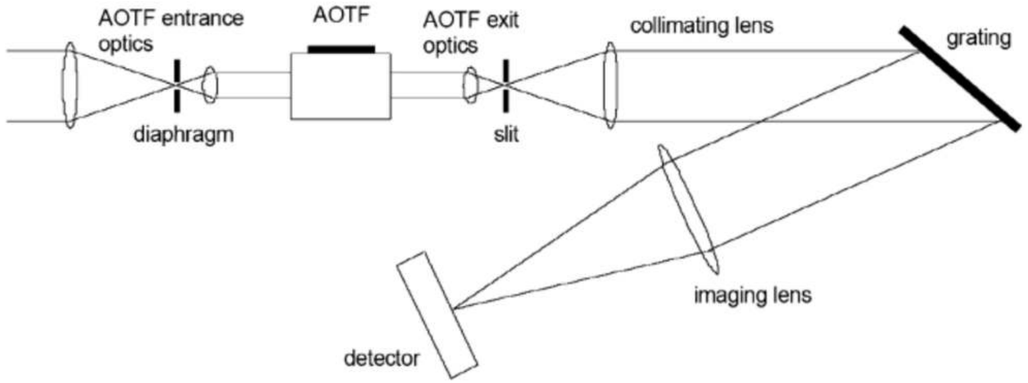


Figure 6: Simplified optical scheme of SOIR (Bertaux et al., 2007)

MECHANICAL SUBSYSTEM

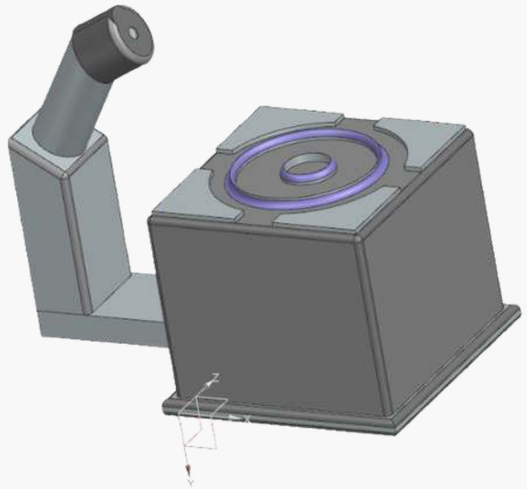


Figure 7: Mechanical Thruster System

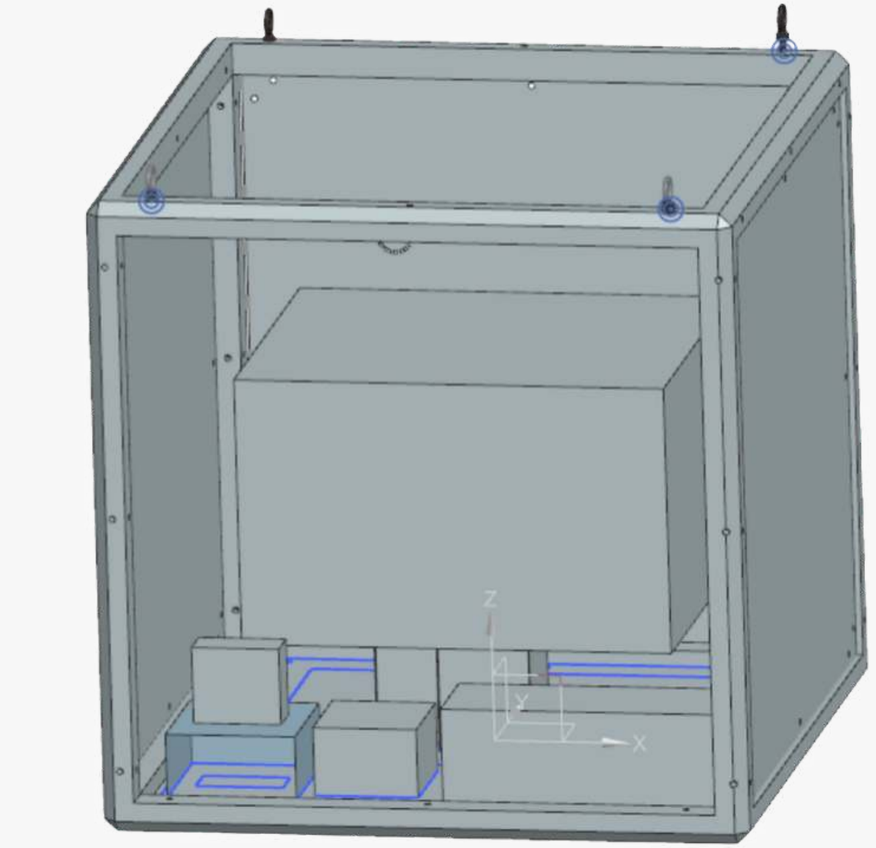


Figure 8: Framework, Chassis, and placeholders for payload and subsystems.

THERMAL SUBSYSTEM

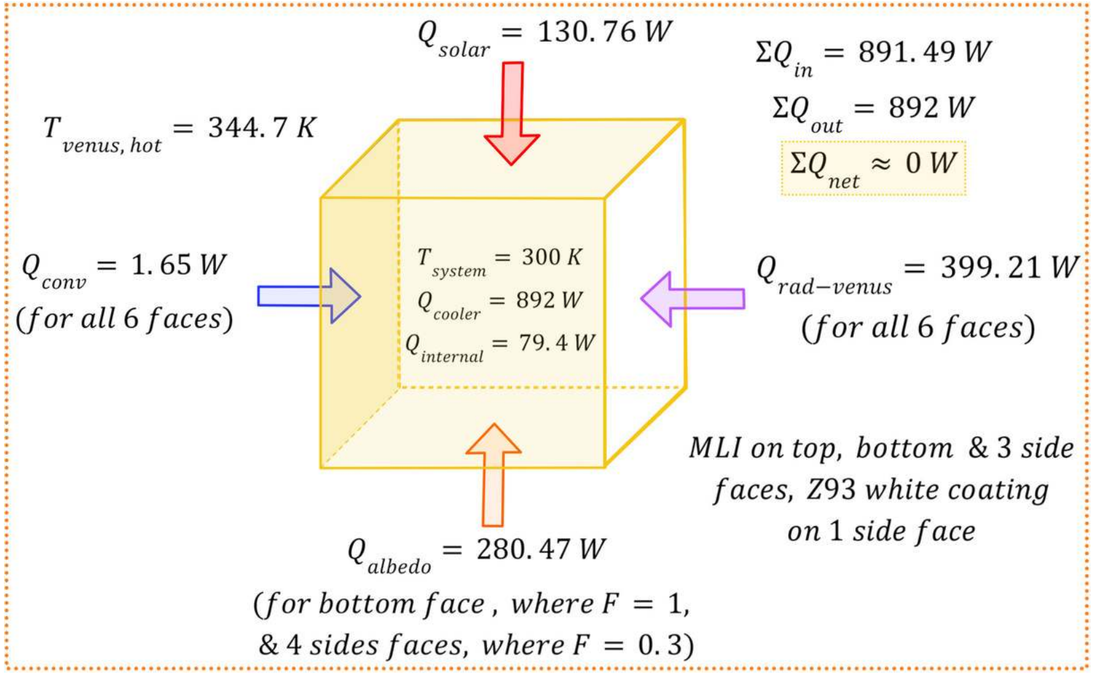


Figure 9: Heat flow map of the hot case (50 km above the surface).

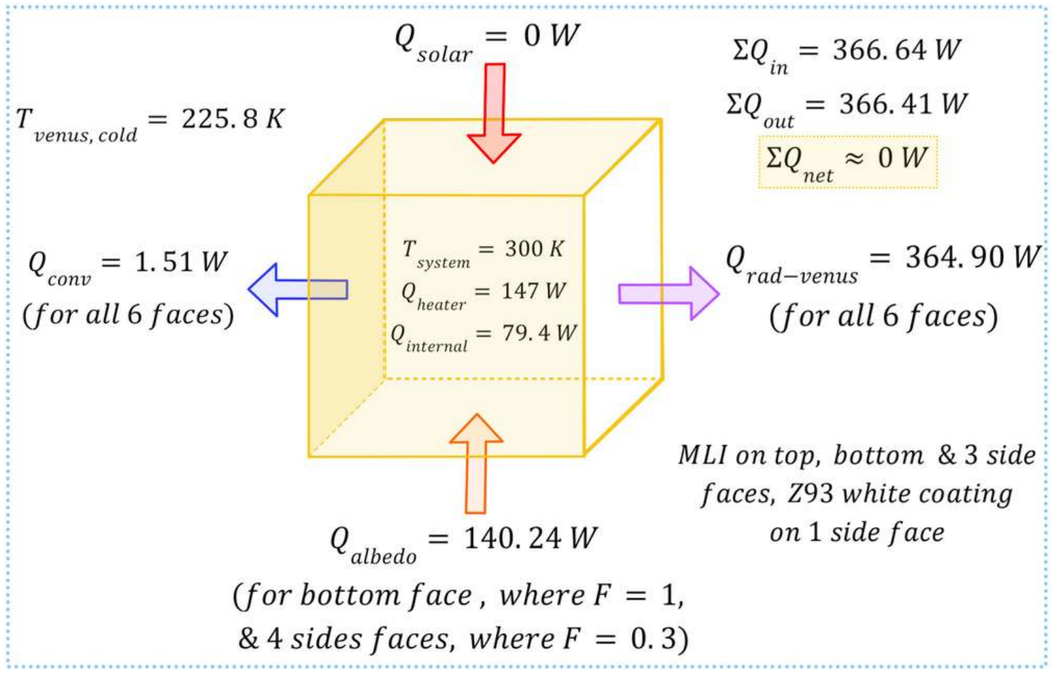


Figure 10: Heat flow map of the cold case (70 km above the surface).

THERMAL SUBSYSTEM

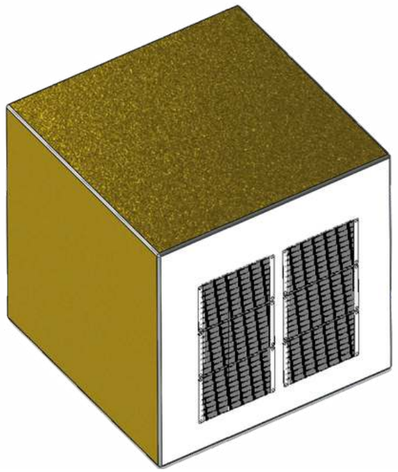


Figure 11: CAD model of the cubic meter system with MLI on five faces and a Z93-coated radiator panel on one face.

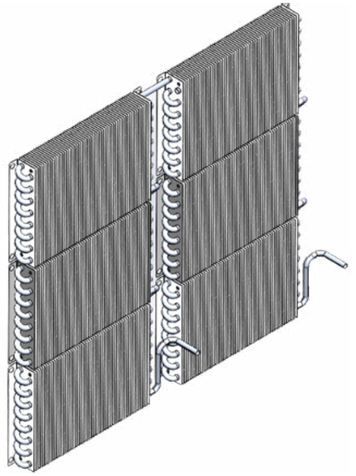


Figure 12: CAD model of the Al 6061 radiator.

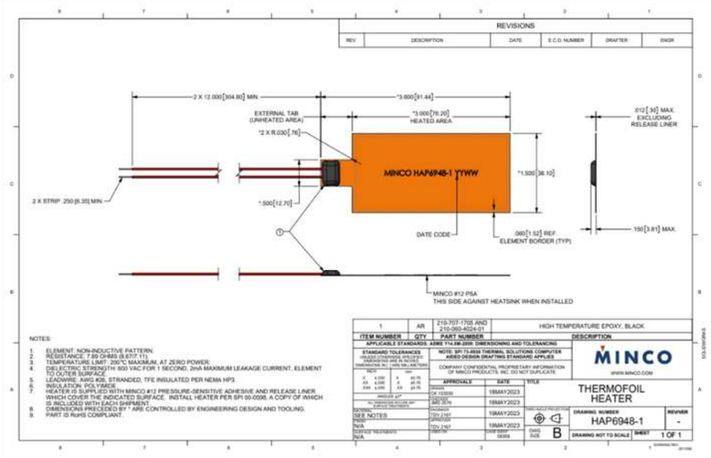


Figure 13: Engineering drawing of the All-Polyimide Thermofoil Heater (HAP6948-1) (Minco, 2023).

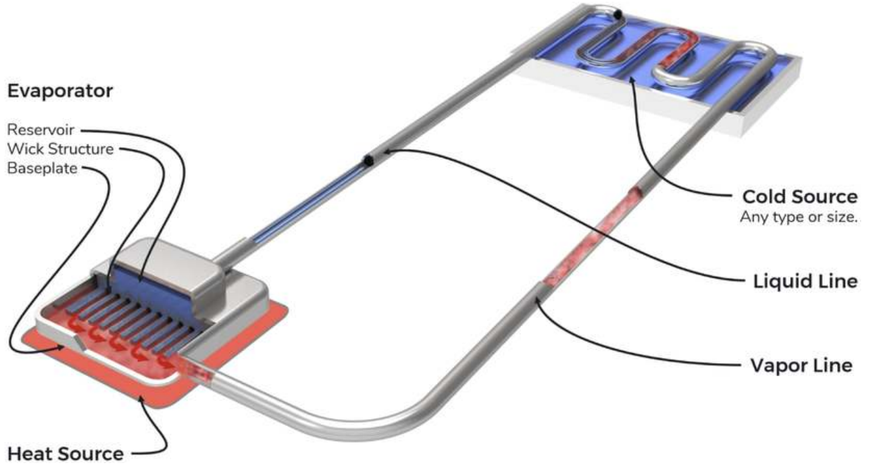
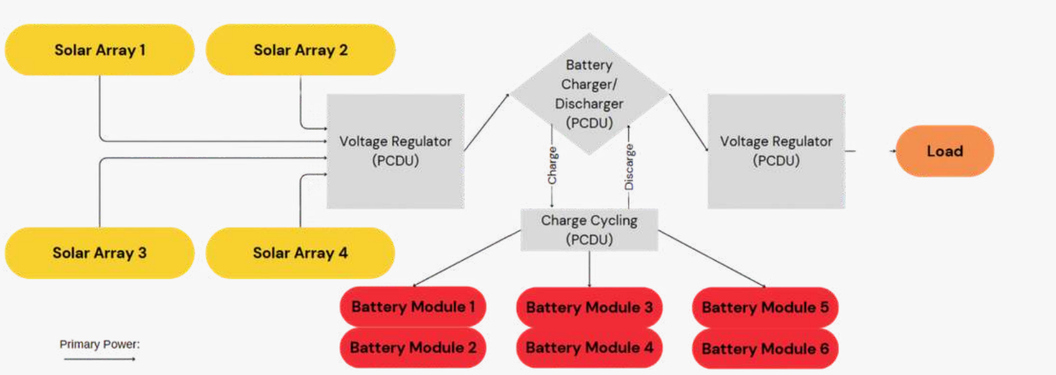


Figure 14: Phase-change cycle and components of a Loop Heat Pipe system (Calyos, 2025).

POWER SUBSYSTEM

Figure 15: This diagram shows the power distribution of the Aerobot System



The Venera Nova Power system is very vital to this mission.

The Power Subsystem includes:

-(4) Solar Arrays

-(6) Thermal Batteries

-(1) Power Conditioning and Distribution Unit

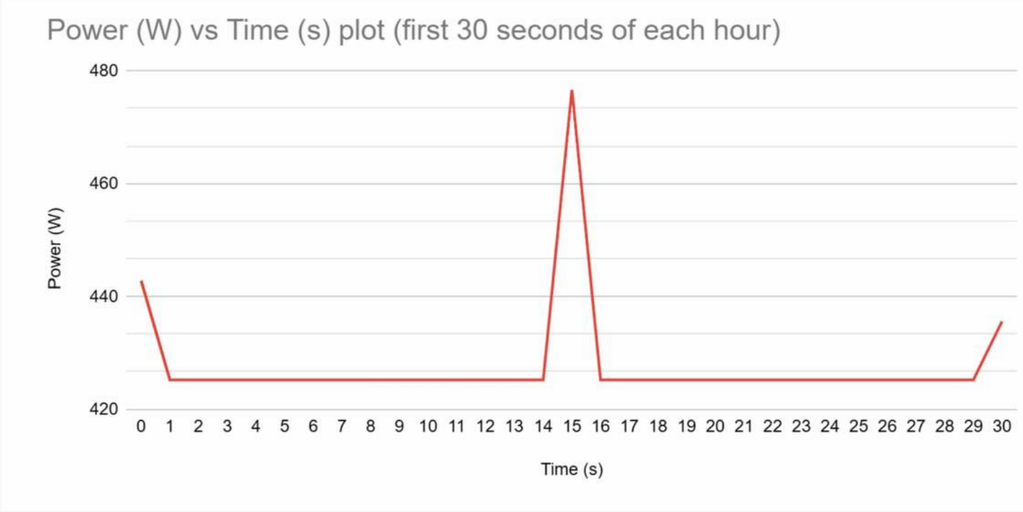


Figure 16: This diagram is a plot of power vs time for the Aerobot System's first 30 seconds of each hour.

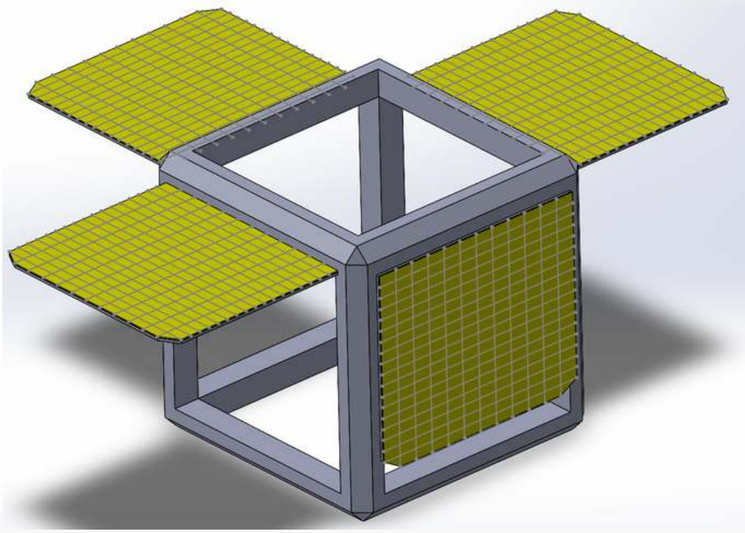


Figure 17: CAD model of the Aerobot's Power Generation Subassembly.

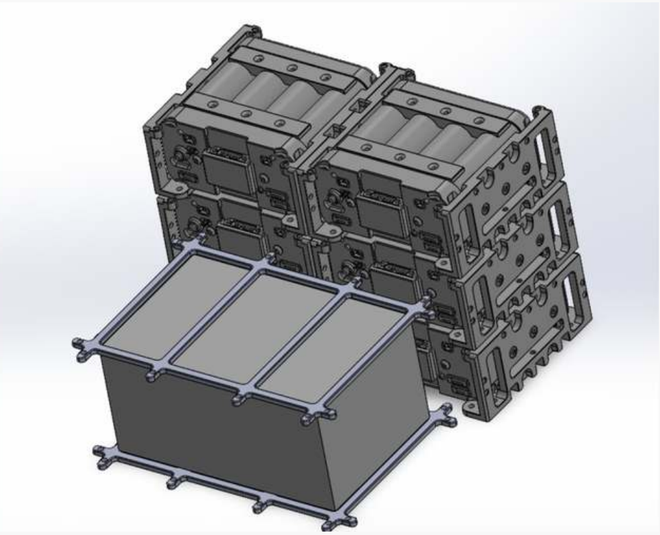


Figure 18: CAD model of the Aerobot's Power Storage and PMAD Subassemblies.

CDH SUBSYSTEM

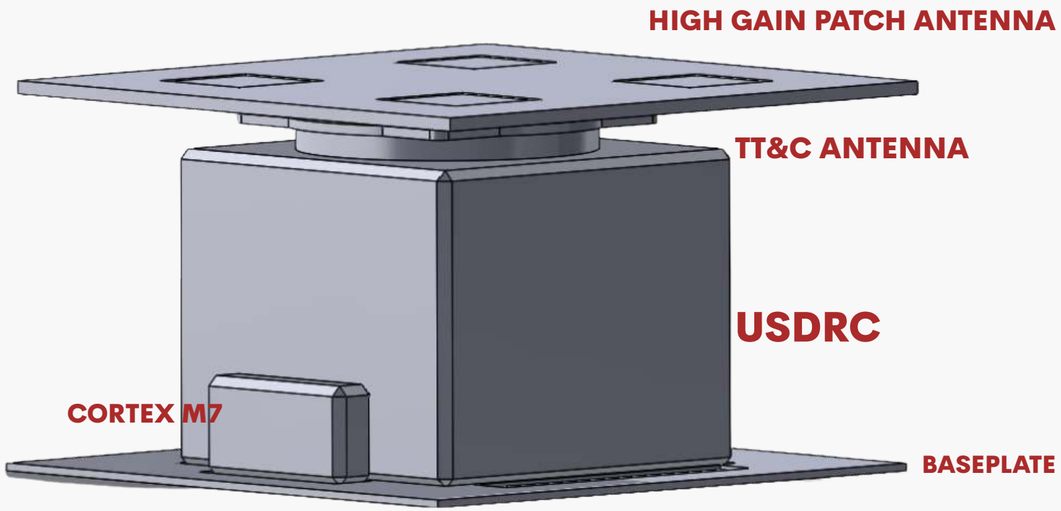
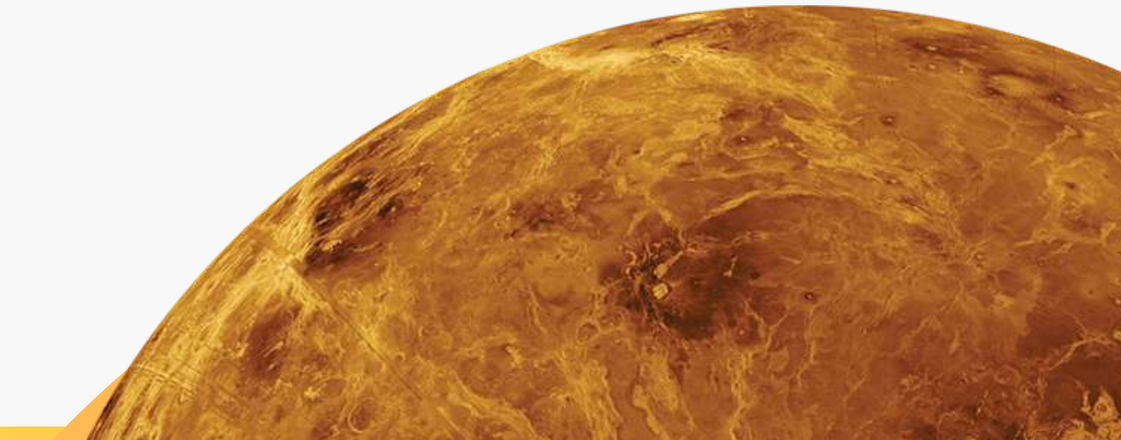


Figure 19: CAD model of the Aerobot's CDH Subsystem

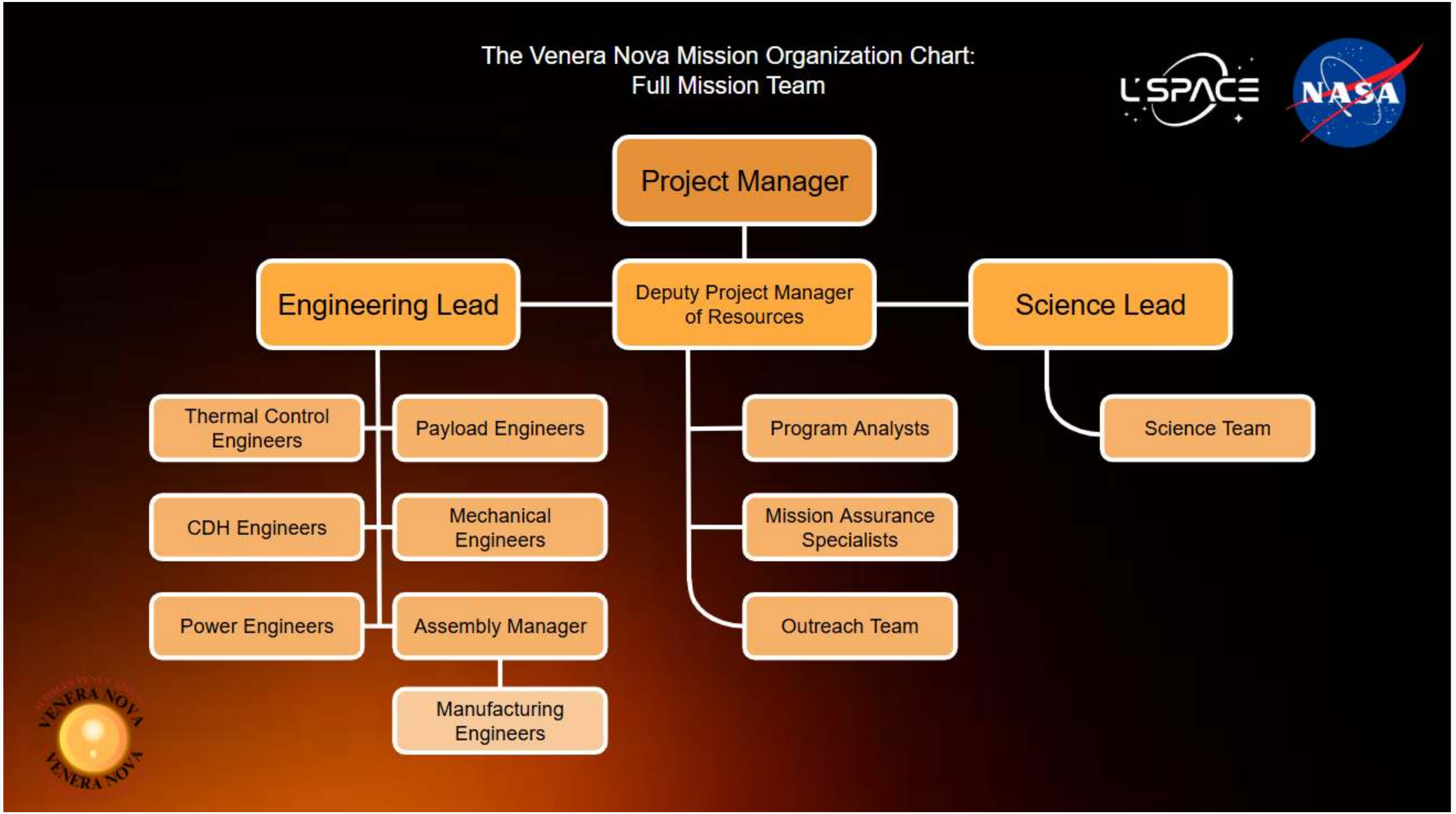


Figure 20: S-Band TT&C Antenna

PROGRAMMATICS



Project Management Approach



Personnel List

	Phase C	Phase C-D	Phase D-E	Phase E-F
Science Personnel	5	5	10	12
Engineering Personnel	12	14	16	18
Technicians	10	10	10	5
Administration Personnel	10	10	10	10
Management Personnel	4	4	4	4

Advanced Risk Log

Rank	Risk ID #	WBS Element	Risk Owner	Category	Timeframe	Risk Title	Risk Statement	L	C	Rating	Approach	Trend	Mitigation Plans	Mitigation Plans	Trigger Date	Trigger Event	Closure Date	Closure Event	Status Updates
1	MECH-2	Spacecraft	Isabella Fernandez Mechanical Engineer	Cost/Scope	Short Term	Aerobot System Depressurization	Given that the Aerobot System shall contain a pressurized environment that is different from the surrounding environment of Venus, there is a possibility that a breach in the Aerobot System will result in a pressure difference and any potential pressure leaks within the Aerobot System. Thus leading to a total mission failure with inoperable systems due to a compromised internal Aerobot environment.	1	1	Moderate	Mitigate (M)	↓ Decrease	1. The internal environment within the Aerobot shall be tested for any pressure leaks prior to takeoff. 2. An ultrasonic leak detector is proved to be quite useful in detecting pressure leaking.	Risk is identified Risk is mitigated	3/10/2025 3/31/2025	SRR MDR	4/21/2025	Project PDR	2/06/2025: Likelihood has been updated from a 4 to 2 after completing mitigation plan 1. 3/31/2025: Likelihood has been updated from 2 to 1 after completing mitigation plan 2.
2	POWER-6	Spacecraft	Andrew Hoang Electrical Engineer	Cost/Scope	Medium Term	Lack of Sufficient Power	Given that the Aerobot System shall contain a necessary amount of stored voltage at every moment of the mission, also enough power input to every mandatory system. There remains a chance that power production/storage/distribution may be compromised. In turn leading to a complete internal system failure.	1	1	Moderate	Mitigate (M)	↓ Decrease	1. Add an extra solar panel for greater power production. 2. Add three extra thermal batteries for backup power.	Risk is identified Risk is mitigated Risk is mitigated	3/06/2025 3/06/2025 3/29/2025	SRR MDR	4/21/2025	Project PDR	3/06/2025: Likelihood has been decreased from a 4 to a 3 after completing mitigation plan 1. 3/29/2025: Likelihood has been updated from a 3 to a 1 after completing mitigation plan 2.
3	THERMAL-11	Spacecraft	Elias Pedroza Thermal Engineer	Cost/Schedule	Long Term	The Overheating of System Components	Given that electronics are highly prone to overheating, there is a possibility that power distribution and system communication is greatly reduced. This can possibly lead to the failure of subsystems and a total mission failure and cancellation. This can lead to the possibility that the spacecraft misses set course entirely.	1	2	Moderate	Mitigate (M)	→ Neutral	1. Implement a thermal cooling system enclosed in a controlled environment.	Risk is identified Risk is mitigated	3/10/2025 3/29/2025	SRR MDR	4/21/2025	Project PDR	3/29/2025: Likelihood has been updated from 4 to 2 after completing mitigation plan 1. 3/31/2025: Risk ID # has been updated from Thermal-11 to CDH-7 4/18/2025: Risk ID # has been reverted from CDH-7 to Thermal-11
4	INSTRUMENTATION-17	Spacecraft	Valerie Miranda Chief Scientist	Cost/Scope	Long Term	VMC (Venus Monitoring Camera) malfunction	Failure of the system in any form would greatly reduce the mission's ability to collect data. Any signal atmosphere would be unable to be imaged and evaluated for atmospheric accelerations.	3	2	Moderate	Mitigate (M)	↓ Decrease	1. Test equipment in similar man-made environments to increase likelihood of survival.	Risk is identified Risk is mitigated	3/10/2025 3/31/2025	SRR MDR	4/21/2025	Project PDR	3/12/2025: Consequences have been updated from 1 to 2 due to more research. 3/31/2025: Likelihood has been updated from 5 to 3 after completing mitigation plan 1. 3/31/2025: INSTRUMENTATION-18 and INSTRUMENTATION-17 have been merged into INSTRUMENTATION-18 to simplify risk.
5	MECH-4	Spacecraft	Elias Pedroza Thermal Engineer	Cost	Short Term	Material Corrosion Exposure	Given the highly corrosive Venusian environment, there is a possibility that corrosion will occur on materials on the outside of the Aerobot System. There is a possibility that the outer shell of the Aerobot System may delaminate and this can lead to the internal destruction of the entire Aerobot System.	5	5	Moderate	Accept (A)	→ Neutral	1. Use material that has a high corrosion resistance. (Only needs to last 48 hours on Venus).	Risk is identified Risk is mitigated Risk is accepted	3/10/2025 3/15/2025 3/31/2025	SRR MDR	4/21/2025	Project PDR	4/15/2025: Consequences have been updated from a 3 to a 5 after completing mitigation plan 1. 3/31/2025: Risk is accepted. 4/21/2025: Risk is updated from THERMAL-12 to MECH-4 due to the material properties inherently being apart of mechanical.
6	THERMAL-12	Spacecraft	Gregory Kimpo-Jabido Systems Engineer	Cost/Schedule	Short Term	Solar Radiation Exposure	Given the extreme solar radiation found in the upper atmosphere of Venus, there is a possibility that other subsystems like CDH, Power, Thermal, etc. can be damaged by solar ray emissions. Data may become corrupted and systems can lose functionality. Thus, the mission can miss out on extremely valuable data, and potentially total system failure.	5	4	Moderate	Accept (A)	→ Neutral	1. Use material that is able to shield internal systems from harmful solar emissions.	Risk is identified Risk is mitigated Risk is accepted	3/06/2025 3/29/2025 3/31/2025	SRR MDR	4/21/2025	Project PDR	3/29/2025: Consequences have been lowered from a 2 to 4 after mitigation plan 1. 3/31/2025: Risk is accepted.
7	PROGRAMMATICS-1	Project Management	Isaac Munoz Mission Assurance Specialist	Cost/Schedule/ Scope	Short Term	Planetary Contamination	Given the arrival of the Aerobot System in the Venusian Atmosphere. There remains possible chances of contamination. The atmosphere of Earth contains small traces of microbial life, and there is a possibility that this life can end up in Venus, using the exterior of the Aerobot System as a shelter. This can potentially affect the environment of Venus.	3	4	Low	Accept (A)	→ Neutral	1. Sterilize the environment within the Aerobot System before mission launch	Risk is identified Risk is accepted	3/25/2025 3/31/2025	SRR MDR	4/21/2025	Project PDR	3/31/2025: Risk is accepted
8	POWER-5	Spacecraft	Andrew Hoang Electrical Engineer	Schedule/Scope	Long Term	Lack of Sunlight	Given the increasing dissipation of sunlight relative to the distance traveled from Venus' upper atmosphere. Solar panels shall lose efficiency due to lack of sunlight. Low power output can result in lower system performance. Overall leading to less data for the Aerobot System can function.	4	3	High	Mitigate (M)	↓ Decrease	1. Add a thermal battery to Aerobot System. 2. Add another solar panel to try and maintain solar power. 3. Add three more batteries to Aerobot System.	Risk is identified Risk is mitigated Risk is mitigated	3/06/2025 3/06/2025 3/29/2025	SRR MDR	4/21/2025	Project PDR	3/06/2025: Consequences are updated from 1 to 2 after mitigation plan 1. 3/29/2025: Consequences are updated from 2 to 3 after mitigation plan 2.
9	PROGRAMMATICS-2	Project Management	Victor Chen Program Analyst	Cost	Long Term	Exceeding Mission Budget	Given the set amount of money granted to the MCA team, there is a chance that all granted money runs out and possibly more is spent than owned. Thus leading to upset stakeholders and quite possibly an early end to the mission.	2	1	Moderate	Mitigate (M)	→ Neutral	1. Track money spent in each deliverable.	Risk is identified Risk is mitigated Mission Descope	3/10/2025 3/10/2025 3/11/2025	SRR MDR Descope	4/21/2025	Project PDR	3/10/2025: Likelihood has been updated from a 3 to 2 after taking mitigation 1 into account

Rank	Risk ID #	WBS Element	Risk Owner	Category	Timeframe	Risk Title	Risk Statement	L	C	Rating	Approach	Trend	Mitigation Plans	Mitigation Plans	Trigger Date	Trigger Event	Closure Date	Closure Event	Status Updates
10	PROGRAMMATICS-3	Project Management	Tyree Nguyen Deputy Project Manager of Resources	Scope	Long Term	Exceeding Mass Constraint	Given the mass constraint set by stakeholder there is a chance that the spacecraft exceeds mass constraints. There is an extremely high chance that this will end the mission if it is not solved before the PDR.	1	1	Moderate	Mitigate (M)	→ Neutral	1. Track mass in each deliverable.	Risk is identified Risk is mitigated Mission Descope	3/10/2025 3/10/2025 3/11/2025	SRR MDR Descope	4/21/2025	Project PDR	3/10/2025: Likelihood is updated from 3 to 2 after mitigation plan 1.
11	MECH-3	Spacecraft	Manahil Syeda Lead Systems Engineer	Scope	Short Term	Dust/Particulate Build Up	Given the high winds and dusty atmosphere of the planet Venus, there is a chance that the Aerobot System shall become covered in dust. This can affect the mass, ventilation, and overall function of the Aerobot System. This shall potentially cut down the Aerobot System's lifetime.	2	3	Low	Accept (A)	↓ Decrease	1. No outside ventilation.	Risk is identified Risk is mitigated	3/10/2025 3/10/2025	SRR MDR	4/21/2025	Project PDR	3/12/2025: Likelihood is updated from a 4 to a 3 after more research of Venus. 3/15/2025: Consequence is updated from a 4 to 5 after mitigation plan 1. 3/15/2025: Likelihood is updated from a 5 to 2 after mitigation plan 1.
12	PAYLOAD-1	Spacecraft	Rachel Brown Project Lead	Scope	Medium Term	Loose Equipment	Any unsecured scientific instruments could potentially be damaged by extreme wind speeds. Intense atmospheric pressure could also damage the scientific instruments and any scientific data collected.	2	4	Low	Mitigate (M)	↓ Decrease	1. Secure equipment to Aerobot System.	Risk is identified Risk is mitigated	3/10/2025 3/10/2025	SRR MDR	4/21/2025	Project PDR	3/15/2025: Likelihood is updated from a 5 to 2 after mitigation plan 1. 3/15/2025: Risk is updated from MECH-1 to PAYLOAD-1 due to this risk being primarily a PAYLOAD problem.
13	THERMAL-14	Spacecraft	Elias Pedroza Thermal Engineer	Cost/Schedule	Medium Term	ATC Heater/Cooler Malfunction	Given the extreme environments of Venus, the Aerobot System shall greatly rely on the cooling function. It is possible that the cooling feature may experience a malfunction. This can lead to overheating and could ruin the entire mission.	1	1	High	Mitigate (M)	↓ Decrease	1. Purchase two ATC coolers 2. Test ATC cooler in a similar scenario in a testing facility	Risk is identified Risk is mitigated Risk is mitigated	3/10/2025 3/31/2025	SRR MDR	4/21/2025	Project PDR	3/31/2025: Likelihood is updated from a 4 to a 3 after mitigation plan 2.
14	PROGRAMMATICS-4	Project Management	Rachel Brown Project Lead	Schedule/Scope	Long Term	Missing Launch Date	Given the strict deadline of March 1st, 2029. the Aerobot System will not be able to be launched any time before or after this date. There is a possibility that the date is missed and the entire mission is called off.	2	1	Moderate	Mitigate (M)	→ Neutral	1. Track schedule in each deliverable	Risk is identified Risk is mitigated Mission Descope	3/10/2025 3/10/2025 3/11/2025	SRR MDR Descope	4/21/2025	Project PDR	3/10/2025: Likelihood is updated from 3 to 2 after mitigation plan 1. 3/11/2025: 6 months must be allocated towards material research, development, and testing. There still may be) less time to advance launch date and PDR. Trend has increased.
15	INSTRUMENTATION-18	Spacecraft	Valerie Miranda Chief Scientist	Cost	Long Term	Solar Flare Exposure	Instrument detectors could be overwhelmed by increased solar radiation due to Venus's weak magnetic field. Solar flares or coronal mass ejections could also interfere with signals, resulting in tarnished data or total data loss.	3	3	Moderate	Mitigate (M)	↓ Decrease	1. Test in similar man-made environment.	Risk is identified Risk is mitigated	3/31/2025	MDR	4/21/2025	Project PDR	3/31/2025: INSTRUMENTATION-18 has been updated to the Risk Title, "Solar Flare Exposure". Risk details have also been updated 3/31/2025: Likelihood has been updated from a 5 to a 3 due to mitigation plan 1.
16	PAYLOAD-8	Spacecraft	Alana Claydio Scientist 1	Scope	Long Term	Environmental Effects on Payload	Environmental hazards such as sulfuric acid clouds, rain, high temperatures, and extreme atmospheric pressures. There is a possibility that payloads on the Aerobot System's subsystem is not appropriately protected. Any subsequent data collected may be insignificant towards accomplishing the mission's scientific objectives.	2	1	High	Mitigate (M)	→ Neutral	1. Test Payload in similar conditions.	Risk is identified Risk is mitigated	3/30/2025 3/31/2025	MDR	4/21/2025	Project PDR	3/31/2025: Likelihood has been updated from 4 to 3 due to mitigation plan 1.
17	POWER-4	Spacecraft	Andrew Hoang Electrical Engineer	Cost/Scope	Long Term	Electrical Components Damaged	Given the many possibilities of internal system failures, there is a possibility that electrical components can have a very vital impact on the success of data gathering and/or Aerobot functionality. Thus having the chance to abort the mission to varying success.	2	2	Low	Accept (A)	↓ Decrease	1. Test electrical components in enclosed environment. 2. Fuses can be implemented to stop excessive current flow.	Risk is identified Risk is mitigated	3/10/2025	SRR	4/21/2025	Project PDR	3/12/2025: Likelihood has been updated from 3 to 1 due to mitigation plan 1.
18	CDH-8	Spacecraft	Gregory Kimpo-Jabido Systems Engineer	Cost/Schedule	Short Term	Signal Interference	Given the high amount of signal interference in space, communication with the Aerobot System and the surrounding Venusian area may be affected. This can lead to system failure and lost data. Thus leading to a mission failure.	2	2	Moderate	Mitigate (M)	↓ Decrease	1. Use circuit analysis code to protect fragile systems	Risk is identified Risks mitigated	3/10/2025 4/20/2025	SRR	4/21/2025	Project PDR	4/20/2025: Likelihood has been updated from a 3 to 2 due to mitigation plan 1.
19	CDH-9	Spacecraft	Victor Chen Program Analyst	Cost/Schedule	Short Term	Cyber-Security Threat	Given the possibility that data can be breached and stolen by hackers, there remains a possibility that data Viruses/tampering can make the data collected obsolete. Functions may also not perform as they are supposed to. Data and systems are highly important to ensuring mission success	4	1	Moderate	Mitigate (M)	↓ Decrease	1. Have a qualified third party conduct a computer vulnerability analysis on project material.	Risk is identified Risk is mitigated	3/10/2025 4/20/2025	SRR	4/21/2025	Project PDR	4/20/2025: Consequences have been updated from a 2 to 5 after more research. 4/20/2025: Likelihood has been updated from a 2 to a 1 after completing mitigation plan 1.
20	CDH-10	Spacecraft	Gregory Kimpo-Jabido Systems	Schedule/Scope	Long Term	Software Error Bugs	Given the chance of possible bugs in CDH software, the programs that shall be implemented during the mission may run into possible failures that keep systems from	1	2	Low	Mitigate (M)	→ Neutral	1. Check code with multiple experts 2. Use successful code from other	Risk is identified Risk is mitigated	3/10/2025 4/20/2025	SRR	4/21/2025	Project PDR	4/20/2025: Likelihood has been updated from 2 to 1 after

WBS Element	Risk Owner	Category	Timeframe	Risk Title	Risk Statement	L	C	Rating	Approach	Trend	Mitigation Plans	Mitigation Plans	Trigger Date	Trigger Event	Closure Date
Spacecraft	Elias Pedroza Thermal Engineer	Cost/Schedule	Short Term	Corrosion	Given Venus's toxic atmosphere, there is a possibility of corrosion of material, resulting in the breakdown of the aerobot, thus failing to safely house equipment needed for mission tasks.	2	1	Moderate	Mitigate (M)	↓ Decrease	1. Check code with multiple experts. 2. Test material used in similar scenario in a testing facility.	Risk is identified Risk is mitigated	3/10/2025 4/20/2025	SRR MDR	4/21/20
Project Management	Manahil Syeda Lead Systems Engineer	Cost/Scheduler/ Scope	Long Term	Balloon Material Degradation	Given the highly corrosive atmosphere of Venus, material degradation is very common. This can be a point of failure that can possible end the mission early. This in turn will limit the amount of data collected.	1	2	Low	Accept (A)	↓ Decrease	1. Use corrosive resistant materials on the Aerobot Balloon. 2. Perform additional material research, development, and testing of the balloon material to fully meet the performance requirement.	Risk is identified Mission Descope	3/10/2025 3/11/2025	SRR Descope	3/11/20
Spacecraft	Rachel Brown Project Lead	Cost/Scheduler/ Scope	Long Term	Equipment Failure Due to Limited Testing with Integration of DSS	Given the DSS system belongs to the DSS contractor, there shall be only a limited amount of time to test and integrate DSS system. Thus likelihood and consequences are unknown until DSS is tested.	3	2	Moderate	Accept (A)	→ Neutral	1. Test DSS system as soon as possible.	Risk is identified Mission Descope	3/11/2025 3/11/2025	Descope MDR	4/21/20
Project Management	Victor Chen Program Analyst	Cost/Schedule	Medium Term	Risk of Exceeding Mission Budget	Given the 20 million dollars that must be allocated to material study and implementation, the MCA Team will run out of resources faster and due to this the team may not be able to afford mitigation strategies or parts for subsystems. Thus, making it more difficult to lower risk.	2	1	Low	Accept (A)	Closed	1. Keep a budget list that is updated bi-weekly.	Risk is identified	3/11/2025	Descope	3/11/20
Project Management	Isaac Munoz Mission Assurance Specialist	Schedule/Scope	Medium Term	Risk of Exceeding Aerobot System Mass	Given the extra 5 kg mass that must be added to the Aerobot System, the MCA team shall have less mass to allocate towards sub-systems this can take away from any backup components which increases the risk.	2	1	Low	Accept (A)	Closed	1. Keep a list of mass that is updated bi-weekly.	Risk is identified	3/11/2025	Descope	3/11/20
Project Management	Isaac Munoz Mission Assurance Specialist	Schedule	Long Term	Risk of Missing Aerobot Launch Date	Given the 6 months that must be added due to the Mission Descope, the MCA Team runs the risk of going over schedule due to 6 months of time-going directly to material study and implementation. Less time to work on the Aerobot System can make planning more difficult and put stress on team members. This can increase the overall risk of the mission.	2	1	Low	Accept (A)	Closed	1.Update calendar according to the loss of the 6 months due to mission descope.	Risk is identified	3/11/2025	Descope	3/11/20

Risk Matrix

Risk Matrix						
Likelihood	5	5→	6→, 7→	8↓		
	4					
	3	11↓			4↓, 22→	13↓, 16→
	2		12↓	15↓	3→, 18↓	9→, 10→, 14→
	1	19↓			17↓, 20→, 21↓	1↓, 2↓
	#	5	4	3	2	1
		Consequences				

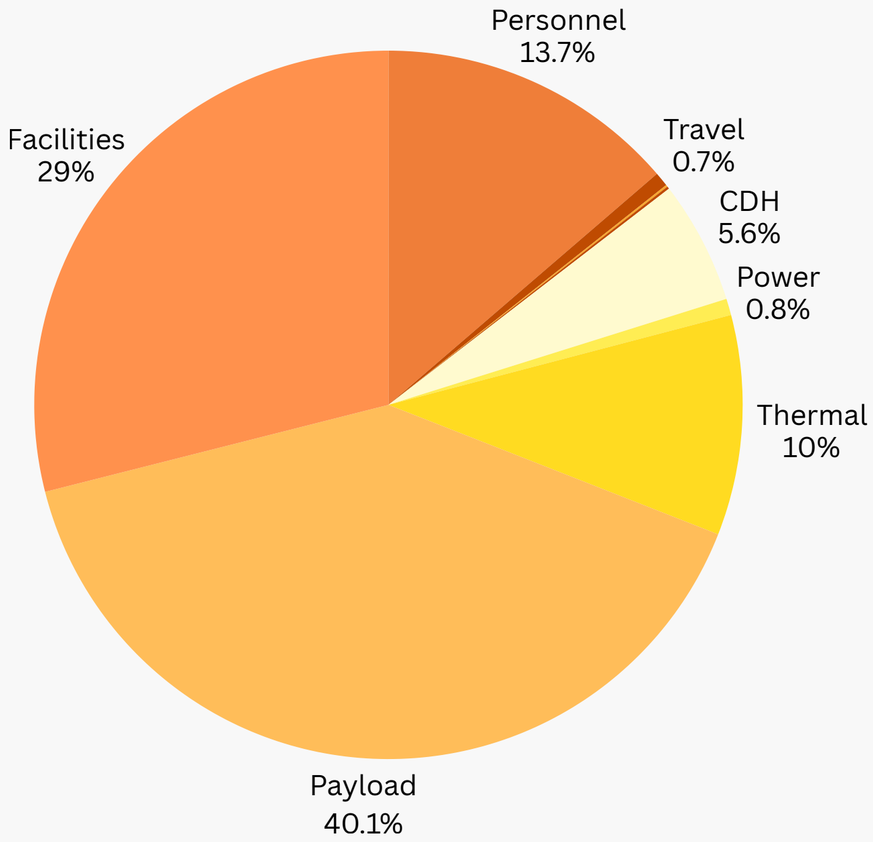
Failure Mode and Effect Analysis (FMEA) Table

Function	Failure Mode (Risk Identification)	Effects	Sev	Cause	Occ	Prevention	Det	RPN	Actions
Power Subsystem	Power-5: Lack of Sunlight	With Solar cells being the Aerobot's primary source of energy, a lack of sunlight could result in not only the Power subsystem but also all other systems from receiving inadequate to no power.	9	Venus' dense atmosphere can cause sunlight to be blocked from reaching the spacecraft.	8	Ensure to research and predict the environment of Venus' atmosphere at the time of the launch date and test backup power generation systems of the overall system to ensure it can survive a lack of sunlight.	2	144	The aerobot's power subsystem does include redundant systems in order to provide stored or redundant energy sources, including backup batteries and generators. This will help provide at least a few more hours of research and data to be collected before the system fully powers down

Personnel Hazards and Mitigations Table

Subsystem	All				
Hazard	Breathing in metal dust and/or welding fumes.	Hearing damage from prolonged exposure to noise.	Injuries from coming into contact with sharp edges.	Injuries from manufacturing at different heights.	Improper storage can result in objects catching fire.
Mitigation Strategy	Provide protective equipment such as masks.	Provide protective equipment such as earplugs.	Implement edge protectors and padding around harsh surfaces. Provide protective gloves to manufacturers.	Ensure workers are properly supervised at all times. Establish physical barriers between the worker and the fall hazard. Provide protective equipment such as full-body harnesses and anchors.	Store all objects outside of the manufacturing space. Only objects pertinent towards manufacturing will be allowed in the area.

Cost Overview



FINAL COST CALCULATIONS					
Total F&A	$ 2,199,274	$ 2,263,585	$ 2,327,895	$ 2,392,206	**$ 9,182,960**
Total Projected Cost	$ 42,438,139	$ 43,867,825	$ 45,891,116	$ 47,196,611	**$ 179,393,691**
Total Cost Margin	$ 5,588,368	$ 5,588,368	$ 5,588,368	$ 5,588,368	**$ 22,353,474**
	13.2%	12.7%	12.2%	11.8%	
Total Project Cost	$ 42,438,139	$ 43,867,825	$ 45,891,116	$ 47,196,611	**$ 179,393,691**

+ $20 million for DSS Descope

= $199,393,691

Personnel Costs

PERSONNEL					
Science Personnel	$ 400,000	$ 410,400	$ 841,600	$ 1,034,880	$ 2,686,880
Engineering Personnel	$ 960,000	$ 1,149,120	$ 1,346,560	$ 1,552,320	$ 5,008,000
Technicians	$ 600,000	$ 615,600	$ 631,200	$ 323,400	$ 2,170,200
Administration Personnel	$ 600,000	$ 615,600	$ 631,200	$ 646,800	$ 2,493,600
Project Management	$ 480,000	$ 492,480	$ 504,960	$ 517,440	$ 1,994,880
Total Salaries	$ 3,040,000	$ 3,283,200	$ 3,955,520	$ 4,074,840	**$ 14,353,560**
Total ERE	$ 848,464	$ 916,341	$ 1,103,986	$ 1,137,288	**$ 4,006,079**
Personnel Margin	$ 513,846	$ 513,846	$ 513,846	$ 513,846	**$ 2,055,384**
TOTAL PERSONNEL	$ 4,402,310	$ 4,835,935	$ 5,863,166	$ 6,172,600	$ 21,274,011

	Phase C	Phase C-D	Phase D-E	Phase E-F
# People on Team	FY 1	FY 2	FY 3	FY 4
Science Personnel:	5	5	10	12
Engineering Personnel:	12	14	16	18
Technicians:	10	10	10	5
Administration Personnel:	10	10	10	10
Management Personnel:	4	4	4	4

Travel Costs

TRAVEL					
Total Flights Cost	$ 75,194	$ 75,194	$ 75,194	$ 75,194	$ 300,776
Total Hotel Cost	$ 22,166	$ 22,166	$ 22,166	$ 22,166	$ 88,664
Total Transportation Cost	$ 20,000	$ 20,000	$ 20,000	$ 20,000	$ 80,000
Total Per Diem Cost	$ 16,056	$ 16,056	$ 16,056	$ 16,056	$ 64,224
Travel Margin	$ 110,895	$ 110,895	$ 110,895	$ 110,895	$ 443,582
Total Travel Costs	$ 244,311	$ 250,663	$ 257,016	$ 263,368	$ 1,015,358

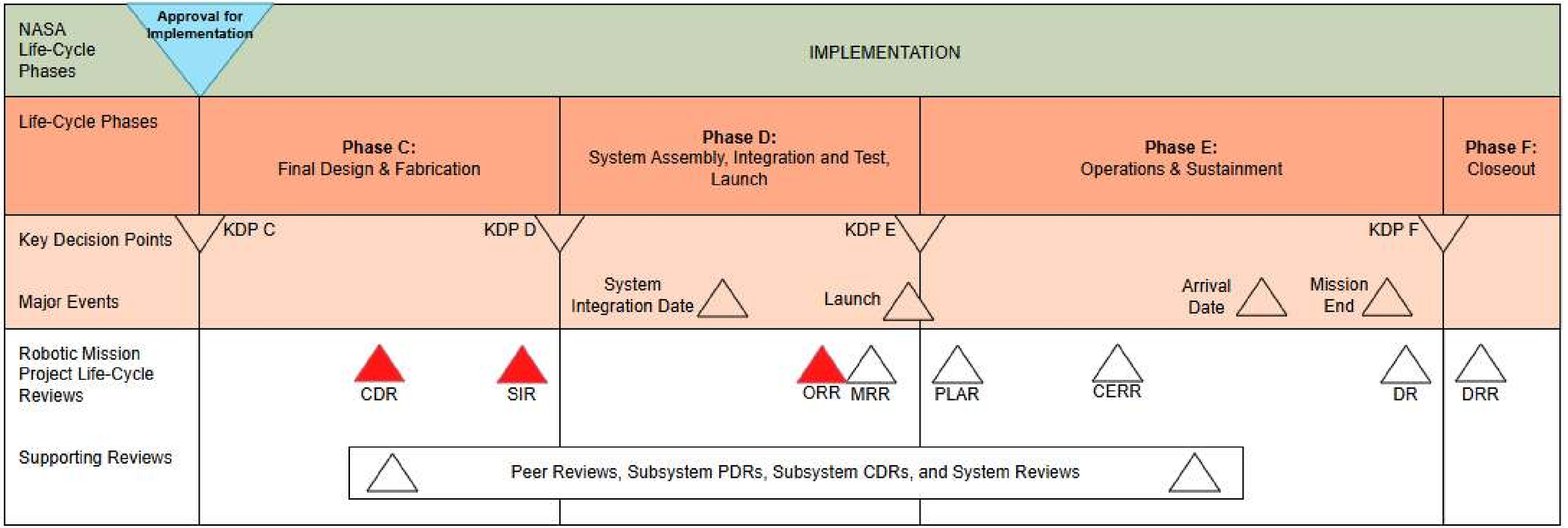
Outreach Costs

OUTREACH					
Total Outreach Materials	$ -	$ -	$ -	$ -	$ -
Total Outreach Venue Costs	$ -	$ -	$ -	$ -	$ -
Total Outreach Travel Costs	$ 1,374	$ 1,374	$ 1,374	$ 1,374	$ 5,496
Total Outreach Services Costs	$ 2,000	$ 2,000	$ 2,000	$ 2,000	$ 8,000
Total Outreach Personnel Costs	$ 15,000	$ 15,000	$ 15,000	$ 15,000	$ 60,000
Outreach Margin	$ 15,272	$ 15,272	$ 15,272	$ 15,272	$ 61,088
Total Outreach Costs	$ 33,646	$ 34,521	$ 35,396	$ 36,270	$ 139,833

Direct Costs

DIRECT COSTS					
Mechanical Subsystem	$ 43,942	$ 43,942	$ 43,942	$ 43,942	$ 175,768
Power Subsystem	$ 291,790	$ 291,790	$ 291,790	$ 291,790	$ 1,167,160
Thermal Control Subsystem	$ 3,900,000	$ 3,900,000	$ 3,900,000	$ 3,900,000	$ 15,600,000
Comms & Data Handling Subsystem	$ 2,182,010	$ 2,182,010	$ 2,182,010	$ 2,182,010	$ 8,728,040
Guidance, Nav, & Control Subsystem	$ -	$ -	$ -	$ -	$ -
Science Instrumentation	$ 15,575,000	$ 15,575,000	$ 15,575,000	$ 15,575,000	$ 62,300,000
Spacecraft Cost Margin	$ 2,742,055	$ 2,742,055	$ 2,742,055	$ 2,742,055	$ 10,968,220
Total Spacecraft Direct Costs	$ 24,734,797	$ 25,377,902	$ 26,021,006	$ 26,664,111	$ 102,797,816
Manufacturing Facility Cost	$ 4,925,000	$ 4,925,000	$ 4,925,000	$ 4,925,000	$ 19,700,000
Test Facility Cost	$ 3,692,500	$ 3,692,500	$ 3,692,500	$ 3,692,500	$ 14,770,000
Facility Cost Margin	$ 2,206,300	$ 2,206,300	$ 2,206,300	$ 2,206,300	$ 8,825,200
Total Facilities Costs	$ 10,823,800	$ 11,105,219	$ 11,386,638	$ 11,668,056	$ 44,983,713
Total Direct Costs	$ 35,558,597	$ 36,483,121	$ 37,407,644	$ 38,332,168	$ 147,781,529
Total MTDC	$ 24,734,797	$ 25,377,902	$ 26,021,006	$ 26,664,111	$ 102,797,816

Schedule Basis of Estimate



FOOTNOTES

1. Flexibility is allowed as to the timing, number, and content of reviews as long as the equivalent information is provided at each KDP and the approach is fully documented in the deliverables.
2. CERR is not necessary, but included as an additional measure to evaluate spacecraft readiness in order to perform critical events/operations, addressing and resolving risks while maintaining spacecraft health and capability.
3. Robotic Mission Project Life-Cycle Reviews was selected due to mission task involving an Aerobot System without onboard humans.
4. Other Reviews were not included due to not being applicable in robotic missions.
5. NASA can request for additional SRBs if deemed necessary.

▲ Red triangles represent life-cycle reviews that require SRBs. The Decision Authority, Administrator, MDAA, or Center Director may request the SRB to conduct other reviews.

ACRONYMS

CDR - Critical Design Review
CERR - Critical Events Readiness Review
DR - Recommissioning Review
DRR - Disposal Readiness Review
KDP - Key Decision Point
MRR - Mission Readiness Review
ORR - Operational Readiness Review
PLAR - Post-Launch Assessment Review
SIR - System Integration Review
SRB - Standing Review Boards

MISSION TIMELINE

PHASE C
5/4/2025

Activities are performed to complete the detailed design of the system and its associated subsystems.

PHASE D
3/4/2026

Instruments and subsystems are constructed and tested before being delivered to Goddard Space Flight Facility.

1/13/2028

The Descent Subsystem (DSS) Contractor is allowed 6 months to deliver the DSS to the Aerobot while other components are still being manufactured and tested.

10/1/2028

System Integration Date where the Aerobot System is integrated with the other components at Goddard Space Flight Facility in Greenbelt, Maryland.

PHASE E
3/1/2029

Venera Nova is launched from Cape Canaveral, Florida. Spacecraft operations and sustainment protocols are initiated.

9/17/2029

Primary Spacecraft enters the Venus orbit and the EDI System is deployed. Mission operations are conducted and spacecraft support is maintained.

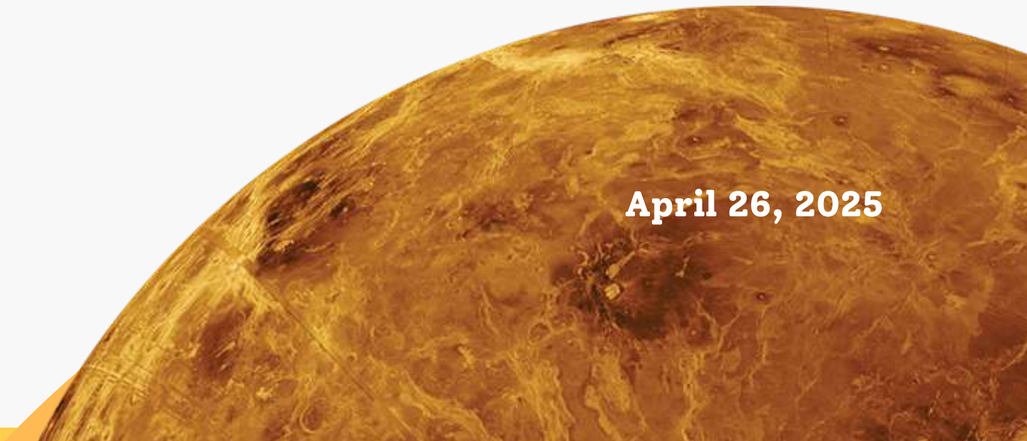
PHASE F
9/19/2029

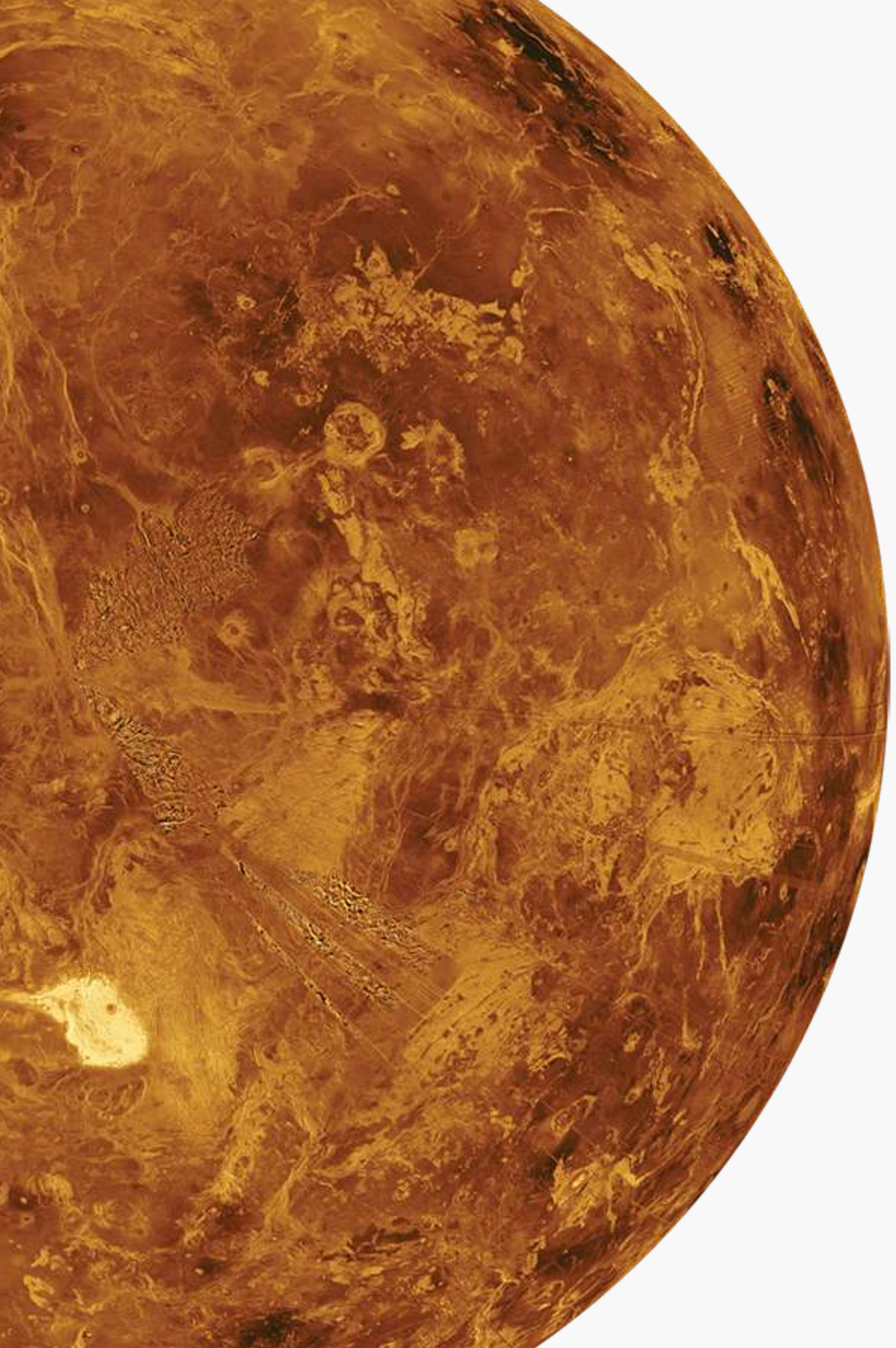
End of Venera Nova mission. Mission results and data collected are analyzed while all technical information is properly identified and archived.

NASA L’SPACE Mission
Concept Academy

THANK YOU

April 26, 2025





AUDACES VENUS ADIUVAT
VENERA NOVA
VENERA NOVA
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